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WS INC

TEAM SLIDE

- Team Name:PixelPirates
- Project Title/Name: Supply Chain - Returns Intelligence
- Members: Ishita Ambekar, Siddhi Vaidya, Shreya Gujarathi, Sanika Shidore

Problem Statement

Returns are not just losses — they are unutilized intelligence

The Problem

- E-commerce companies face a major hidden challenge:
- Customers return products with vague reasons like “does not match description”
- Real causes are buried in unstructured data (reviews, notes, support chats)
- Data exists across disconnected systems with no unified intelligence layer
- No mechanism to connect signals and generate actionable insights

Example:

- A dining table appears warm brown online but looks grey in real lighting.
- Customer selects “does not match description.”
- The company incurs losses, but the actual issue (color mismatch) is never identified.

Why It Matters:

1. Financial Loss
Each return incurs shipping, reverse logistics, and processing costs
A ₹50,000 product can lead to ₹5,000–₹10,000 loss per return
Scales to significant revenue loss across operations

2. Lack of Visibility
High return rates are visible, but root causes are not
Teams cannot identify what to fix or improve

3. No Feedback Loop
Product teams receive delayed or no actionable insights
Issues are addressed only after substantial losses

4. Regional Blind Spots
Return patterns vary by region
Lack of analysis leads to missed localized insights

5. Customer Experience Impact
Mismatch between online representation and real-world product
Leads to dissatisfaction, negative reviews, and brand damage

6. Lost Intelligence Opportunity
Each return contains valuable insights
Currently, this data is not leveraged for decision-making

Solution Overview

High-Level Solution

We built a Returns Intelligence Engine that transforms fragmented return data into actionable business insights.

- Integrates data from returns, reviews, and support systems
- Uses AI/ML to identify root causes, risks, and losses
- Provides real-time alerts and recommendations

Enables teams to move from reactive handling → proactive decision-making

AI Assisted Problem Solving

AI-Assisted Problem Solving

How AI Helped Break Down the Problem

- Converted unstructured data (returns, reviews, support chats) into structured insights
- Identified hidden patterns using clustering and sentiment analysis
- Enabled decomposition of the problem into:
  - Root cause detection
  - Return risk prediction
  - Loss forecasting
- Helped shift from reactive analysis → proactive decision-making

Iterations & Learning

- Iteration 1: Basic keyword-based rules → limited accuracy
- Iteration 2: Added sentiment analysis → improved context understanding
- Iteration 3: Introduced clustering (TF-IDF + KMeans) → dynamic root cause detection
- Iteration 4: Built ML models (ensemble + regression) → risk and loss prediction
- Iteration 5: Integrated LLM layer → better interpretation and summaries
- Iteration 6: Integrated frontend - React, Tailwind CSS, Vite
- Iteration 7: NRV engine and disposition classification
- Iteration 8: Routing - Warehouse and Logistics Optimization - AI informed routing

AI Engineering – Tools Used, Prompt Strategy & Error Handling

1. Tools & Technologies

Data & Backend
Flask (API layer), Pandas (data processing), CSV-based ingestion
Real-time pipeline with caching and refresh

Classical NLP & ML
VADER Sentiment Analysis (text polarity)
TF-IDF + KMeans (root-cause clustering)
VotingClassifier (return risk prediction)
HistGradientBoostingRegressor (loss forecasting)

AI / LLM Layer
LLM APIs (Gemini / Groq) for condition grading
Multimodal AI (Gemini 1.5 Flash) for visual mismatch detection

System Design
Modular pipeline (ingestion → processing → ML → insights)
Dashboard-ready API outputs

Frontend
React 18 (TypeScript) with Vite
react-router-dom
Tailwind CSS (PostCSS + Autoprefixer).

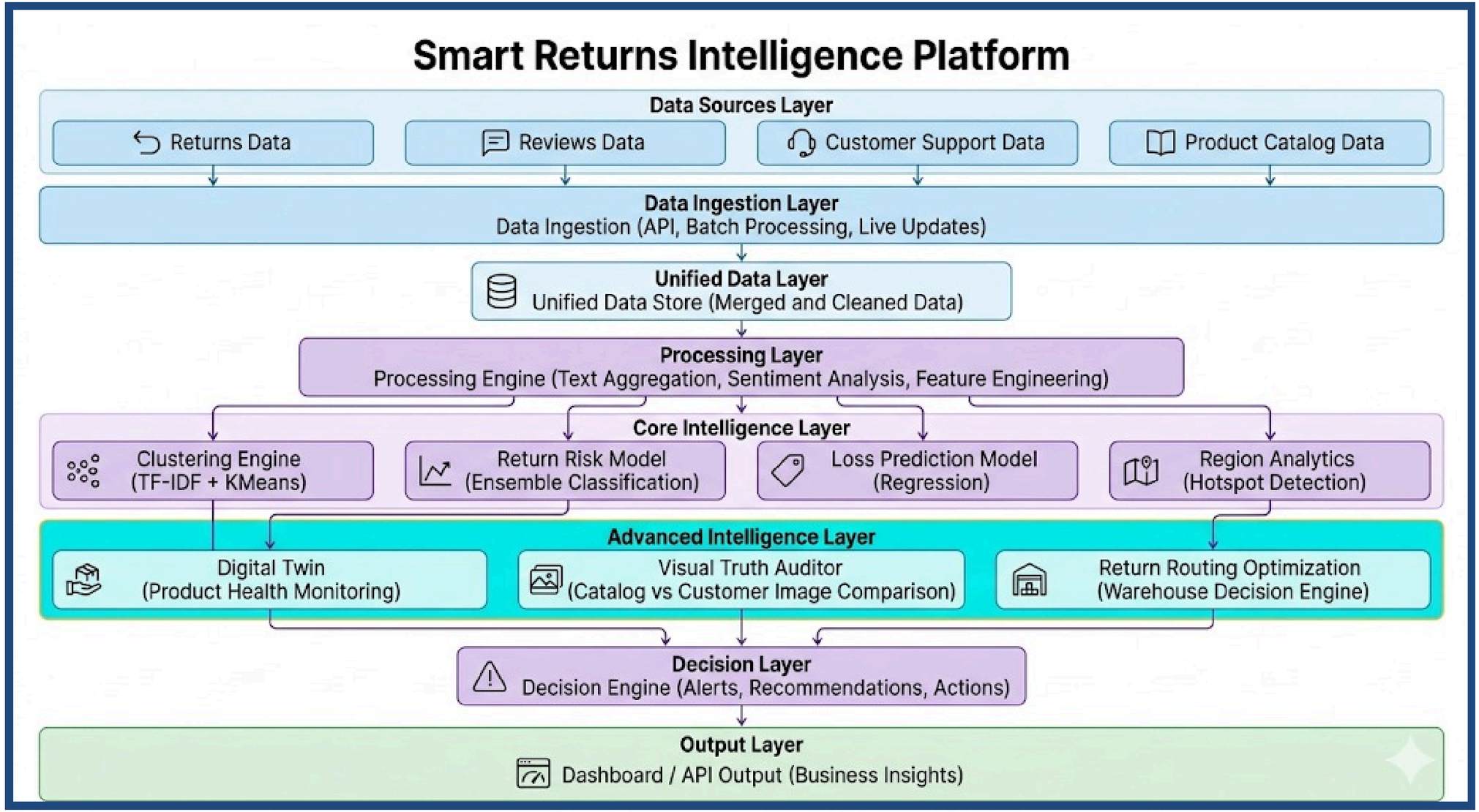
2. Prompt Strategy (for LLM Modules)

Structured prompts
Clear classification instructions (New / Damaged / Faulty)
Context-aware input combining:
Return notes
Customer support transcripts
Low temperature (0.2) for consistent, deterministic outputs
Cluster-based prompting (analyze grouped data instead of per row) for:
Faster performance
Reduced API calls

3. Hallucination & Error Handling

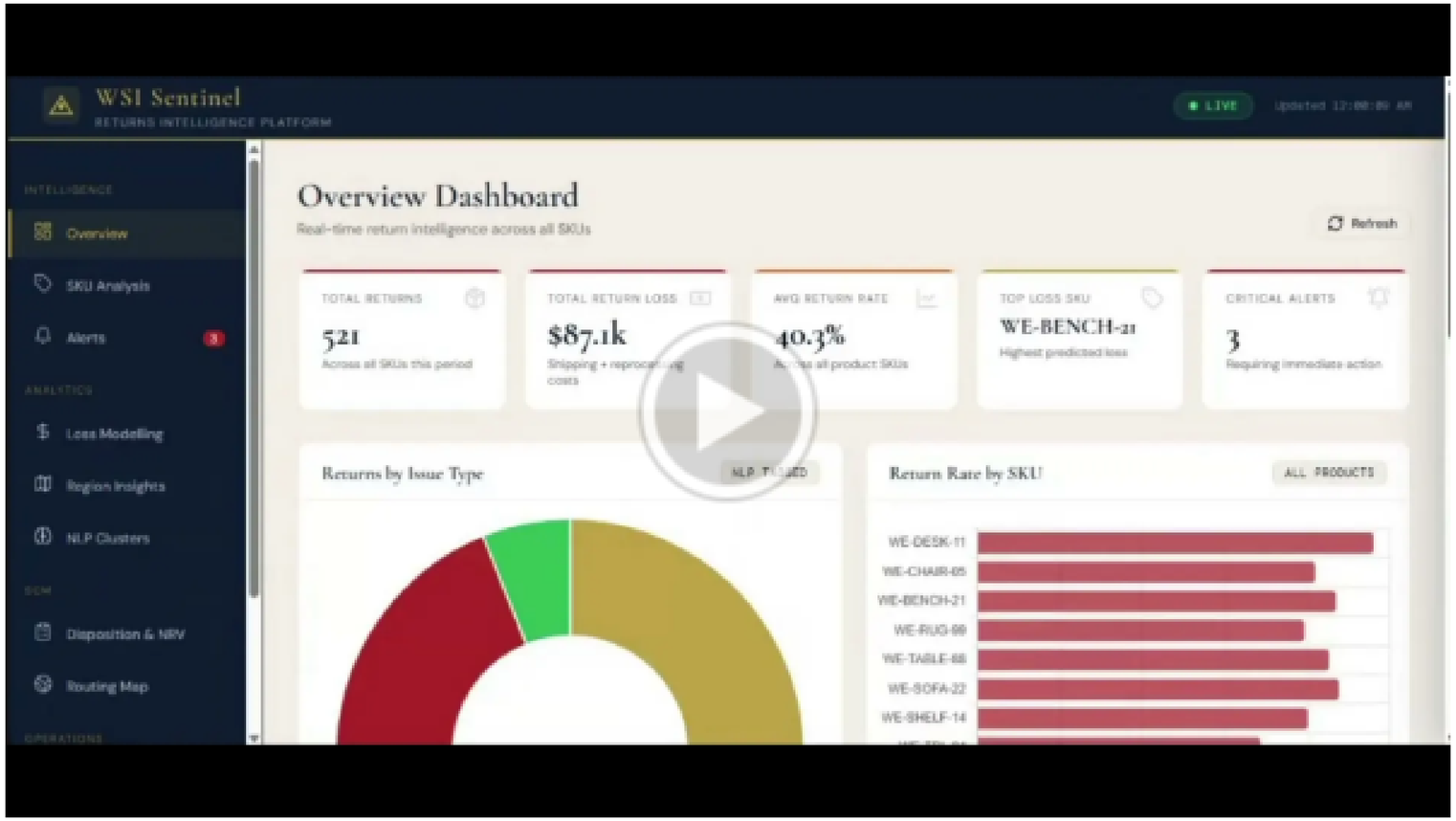
Fallback System If LLM fails or quota exceeds → switch to rule-based logic
Validation LayerEnforce JSON parsing and schema checks
Default values for missing/invalid outputs
Hybrid AI ApproachCritical computations (loss, risk, NRV) remain deterministic
LLM used only for interpretation (non-critical layer)
Rate Limit & Failure HandlingAsync processing with concurrency control
Graceful degradation (system continues without AI dependency)

Technical Architecture



Prototype/Demo

https://drive.google.com/file/d/1M53gPMFb0IPiQsmCM6yeVRmp94rjEHk7/view



Challenges & Learning

1. Challenges in Using AI

- Unstructured & Noisy Data - Return notes, reviews, and transcripts were inconsistent and difficult to analyze
- Vague Labels - Dropdown reasons (e.g., “does not match description”) lacked true root cause
- Small / Synthetic Dataset- Limited real-world data reduced model generalization
- API Limitations - LLM quota errors, latency, and dependency risks
- Overfitting & Bias - Early models gave repetitive outputs (same issue across SKUs)

2. How We Overcame Them

Data Standardization - Combined multi-source text into a unified pipeline
Hybrid AI Approach, Classical ML for risk & loss (reliable, fast), LLMs only for interpretation (non-critical layer)
Clustering Instead of Hardcoding - Used TF-IDF + KMeans to dynamically detect root causes
Fallback Mechanisms - Rule-based system when API fails → ensures reliability
Feature Engineering - Improved dataset with SKU, region, sentiment, and return patterns

3. Future Improvements (Unlimited Resources)

- Real-Time Production Integration

Direct integration with e-commerce platforms for live monitoring

- Advanced Multimodal AI

Compare product images vs real-world images for visual mismatch detection on large scale

- Train deep learning models on millions of real return records

Causal AI & Recommendation Engine

- Large-Scale Training Data

Automatically suggest fixes (image updates, description changes)

- Personalized Return Prediction

Predict returns at user level before purchase

- Global Supply Chain Optimization

Region-aware logistics and dynamic inventory decisions

- Sustainability Twin

We built a scalable, hybrid AI system that transforms returns from a cost center into a decision intelligence engine.

Feedback

Reflections:

This problem highlighted how returns are not just operational issues, but data intelligence gaps
Building this solution helped us understand the importance of combining business context with AI/ML models
We focused on creating a system that is not just predictive, but also actionable and scalable in real-world scenarios
The hybrid approach (ML + LLM + rules) ensures both accuracy and reliability in production environments

Suggestions:

Provide sample real-world datasets or slightly larger datasets for better model training

Experience with the WSI AI-thon

The problem statement was highly relevant and industry-focused, encouraging practical thinking
It pushed us to go beyond basic models and design a complete end-to-end system
The experience was valuable in understanding real-world constraints like data quality, latency, and scalability
Overall, it was a great learning experience combining AI, product thinking, and system design

The AI-thon was an excellent platform to apply AI to real business challenges, and we look forward to participating again.